```
sysadmin@localhost:~$ lscpu
Architecture:        x86_64
CPU op-mode(s):      32-bit, 64-bit
Byte Order:          Little Endian
CPU(s):              16
On-line CPU(s) list: 0-15
Thread(s) per core:  1
Core(s) per socket:  1
Socket(s):           16
NUMA node(s):        1
Vendor ID:           AuthenticAMD
CPU family:          25
Model:               1
Model name:          AMD EPYC 7443P 24-Core Processor
Stepping:            1
CPU MHz:             2844.656
BogoMIPS:            5689.31
Virtualization:      AMD-V
Hypervisor vendor:   VMware
Virtualization type: full
L1d cache:           32K
L1i cache:           32K
L2 cache:            512K
L3 cache:            32768K
NUMA node0 CPU(s):   0-15
Flags:               fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cm
ov pat pse36 clflush mmx fxsr sse sse2 syscall nx mmxext fxsr_opt pdpe1gb rdtscp
 lm constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid tsc_kn
own_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes
 xsave avx f16c rdrand hypervisor lahf_lm svm extapic cr8_legacy abm sse4a misal
ignsse 3dnowprefetch osvw topoext invpcid_single ibpb vmmcall fsgsbase bmi1 avx2
 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec x
getbv1 xsaves clzero wbnoinvd arat npt svm_lock nrip_save vmcb_clean flushbyasid
 decodeassists umip pku ospke vaes vpclmulqdq rdpid overflow_recov succor fsrm
sysadmin@localhost:~$
```

To determine the type of CPU execute the `lscpu` command:

For even more details about your CPU(s), you can examine the `/proc/cpuinfo` file, especially the "flags" that are listed which determine whether or not your CPU has certain features.

Use the `head` command with the `-n` option to list the first 20 lines of the `cpuinfo` file:

```
sysadmin@localhost:~$ free -m
              total        used        free      shared  buff/cache   available
Mem:          64298       12937        5837          26       45523       50616
Swap:          8191           0        8191
sysadmin@localhost:~$ free -g
              total        used        free      shared  buff/cache   available
Mem:             62          12           5           0          44          49
Swap:             7           0           7
```

To discover how much RAM and swap space is being used, use the `free` command

In the output above, you can see that the system has `64298` megabytes (roughly 64 gigabytes) of physical memory (RAM). Of that, only `12937` megabytes are being used, a good sign that you have enough memory for your system's needs.

```
sysadmin@localhost:~$ lspci
00:00.0 Host bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX Host bridge (
rev 01)
00:01.0 PCI bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX AGP bridge (re
v 01)
00:07.0 ISA bridge: Intel Corporation 82371AB/EB/MB PIIX4 ISA (rev 08)
00:07.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
00:07.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:07.7 System peripheral: VMware Virtual Machine Communication Interface (rev 1
0)
00:0f.0 VGA compatible controller: VMware SVGA II Adapter
00:10.0 SCSI storage controller: LSI Logic / Symbios Logic 53c1030 PCI-X Fusion-
MPT Dual Ultra320 SCSI (rev 01)
00:11.0 PCI bridge: VMware PCI bridge (rev 02)
00:15.0 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.1 PCI bridge: VMware PCI Express Root Port (rev 01)
...
```

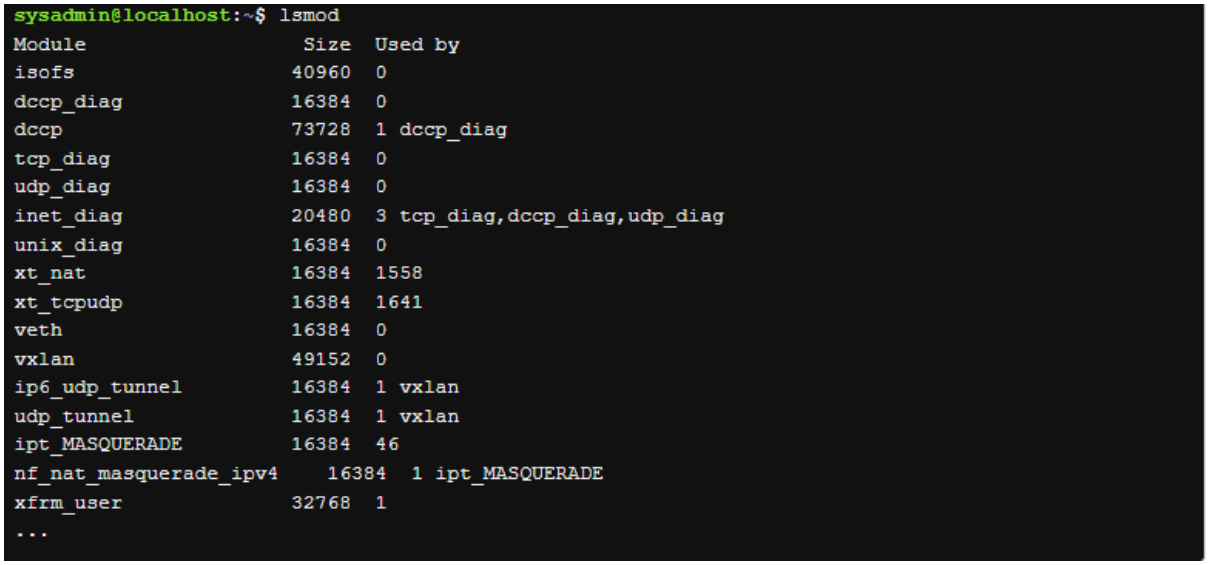
To see what devices are connected to the PCI bus, use the `lspci` command:

```
sysadmin@localhost:~$ lspci -k
00:00.0 Host bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX Host bridge (
rev 01)
        Subsystem: VMware Virtual Machine Chipset
        Kernel driver in use: agpgart-intel
lspci: Unable to load libkmod resources: error -12
00:01.0 PCI bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX AGP bridge (re
v 01)
00:07.0 ISA bridge: Intel Corporation 82371AB/EB/MB PIIX4 ISA (rev 08)
        Subsystem: VMware Virtual Machine Chipset
00:07.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
        Subsystem: VMware Virtual Machine Chipset
        Kernel driver in use: ata_piix
00:07.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
        Subsystem: VMware Virtual Machine Chipset
00:07.7 System peripheral: VMware Virtual Machine Communication Interface (rev 1
0)
        Subsystem: VMware Virtual Machine Communication Interface
        Kernel driver in use: vmw_vmci
00:0f.0 VGA compatible controller: VMware SVGA II Adapter
        Subsystem: VMware SVGA II Adapter
        Kernel driver in use: vmwgfx
...
```

Use the `lspci` command with the `-k` option to show devices along with the kernel driver and modules used:

```
sysadmin@localhost:~$ lsusb
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 001 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
sysadmin@localhost:~$
```

Attempt to list the USB connected devices



Use the `lsmod` command to view the currently loaded modules: